

7 (a) Isolated mitochondria were used to investigate respiration.

- Mitochondria were extracted from respiring mammalian cells and incubated in a buffer solution.
- Pyruvate and inorganic phosphate (P_i) were added at time zero.
- ADP was added one minute later.
- The oxygen concentration of the buffer solution containing mitochondria was monitored throughout the investigation.

The results of the investigation are shown in Fig. 7.1.

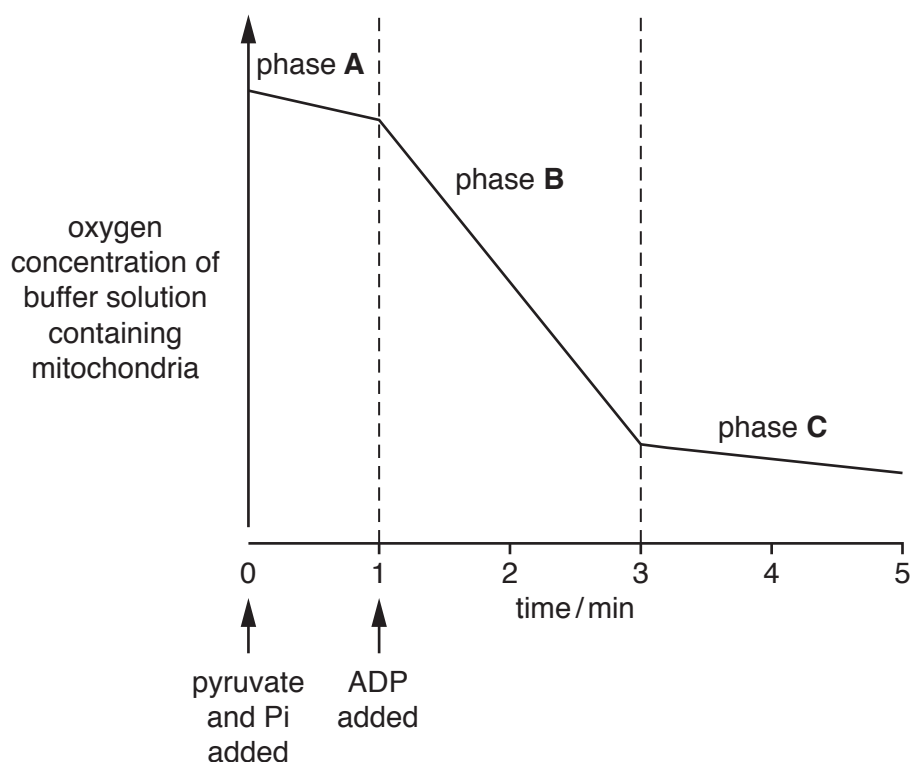


Fig. 7.1

- (i) Suggest why the line of the graph in Fig. 7.1 is steeper during phase **B** than during phase **A**.

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..... [3]

- (ii) Suggest why the line of the graph starts to level out in phase **C**.

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..... [1]

- (b) Some plants, such as rice, grow with their roots submerged in water.

Fig. 7.2 shows a group of rice plants.



Fig. 7.2

Explain how rice is adapted to grow with its roots submerged in water.

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